Assignment Day – 3

1. **Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.**

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Feature	**DDR3**	**DDR4**	**DDR5**
Release Year	2007	2012	2020
Speed	Up to 17 GB/s	Up to 25 Gb/s	Up to 67 GB/s or higher
Voltage	1.5V	1.2V	1.0V
Power consumption	Higher	Lower then DDR3	Lower and most power-efficient
Performance	Good for basic computing	Faster and batter for multitasking and gaming	Fastest, ideal for high-performance computing
Typical use cases	Older desktop and laptop, office work, web browsing	Modern desktop, laptop, gaming PC's	High-end gaming, 3D rendering, video editing and AI
Compatibility	Works only with DDR3 motherboards	Works only with DDR4 motherboards	Works only with DDR5 motherboards

2. **Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully**

   Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

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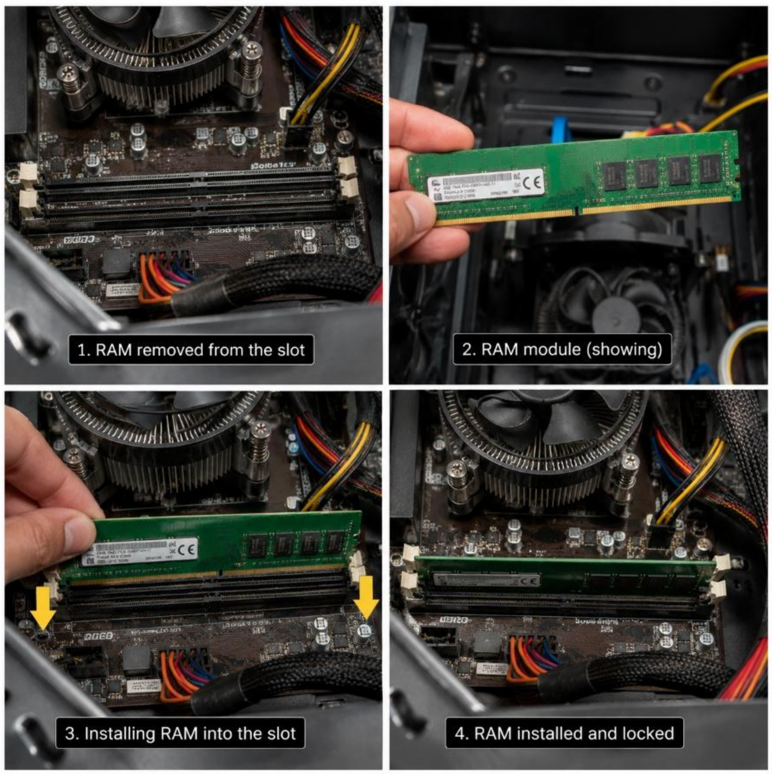

❖ **Steps for Remove and Reinstall**

1. **Turn off the computer.**
2. **Unplug the power cable (and remove the laptop battery if possible).**
3. **Open the computer case or laptop back cover.**
4. **Find the RAM slot.**
5. **Take a photo of the installed RAM.**
6. **Push the side clips to remove the RAM.**
7. **Reinsert the RAM into the same slot until it clicks into place.**
8. **Close the case and reconnect the power.**
9. **Turn on the computer and check that it boots normally.**

3. **List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.**

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Feature	HDD	SSD	NVMe
Speed	Slow (80-160 MB/s)	Fast (500-550 MB/s)	Very Fast (2000-7000+ MB/s)
Storage Technology	Uses Spinning magnetic disks	Uses flash memory (no any moving part)	Uses flash memory
Best use case / Daily App	Photos, Movies and Music – Good for storing large files because it offers high capacity at a low cost	Microsoft Word or Google Chrome – Opens application and files much faster than an HDD	Adobe premiere Pro or Modern Games (e.g. GTA V, Valorant) – Loads large project and games much faster with minimal waiting time

4. **Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly. Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.**

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